

LECTURE - 14

Thursday
9/10/25

Language.

① Reg. Expression

② FA → Reg. lang.

Non. Reg. lang.

→ DFA
→ NFA
→ NFA NULL

CFL

→ CFL

→ PDA (FA + Memory)

Stack

diagram

Dynamic

STACK: using Linked List

→ already top of stack element

a, Z₀ / a Z₀

top 2 elements

input:

about to be added

TOS → Z₀

→ Delimiter:
if only Z₀ in
stack, stack
is empty.



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⇒ PUSH

Inserting 'b' in stack that has 'a' and Z_0 .⇒ POP that is coming in input of FA
input, TOS / ^

• Incoming input is consumed

a, b / ^ : input: 'a' is consumed and then 'b' is popped out of Stack.

• Incoming input is not consumed.

^, b / ^ : b is popped without any input getting consumed.

⇒ Pushing data without consuming input:

Q: Input = ab; add 'aa' whenever there is input 'a'.

a, Z₀ / aZ₀

// 'a' pushed in stack

Syntax

^, a / a^

// since first 'a' pushed; ∴ we'll push another 'a' that we're to put ourself.

about to be added
already pushed TOS in stack. This has to be done without consuming next input i.e. 'b'

∴ we used ^, a not ba.

b, a / ba

// 'b' pushed after two 'a's.

$$FA = (Q, \Sigma, q_0, A, \delta)$$

$$Stack = (\Gamma, Z_0)$$

$\rightarrow$ stack element
 $\rightarrow$ delimiter.

$$PDA = (Q, \Sigma, \Gamma, q_0, A, Z_0, \delta)$$

transition func. of PDA

$$\delta: Q \times \{\Sigma \cup \Lambda\} \times \Gamma \rightarrow Q \times \Gamma^*$$

current state current input TOS element next state of PDA after changing Q, Σ and Γ

eg: single/multiple push or pop, no action etc.

since 'c' is pushed here but is not written in the top two element of stack (after slash) $\therefore$ it means 'c' isn't added in stack. as 'c' is consumed but not added.

$$Q: L = \{n \mid n \in \Sigma^* \text{ \& } n = ycy^r \text{ where } y \in \{a,b\}^*\}$$

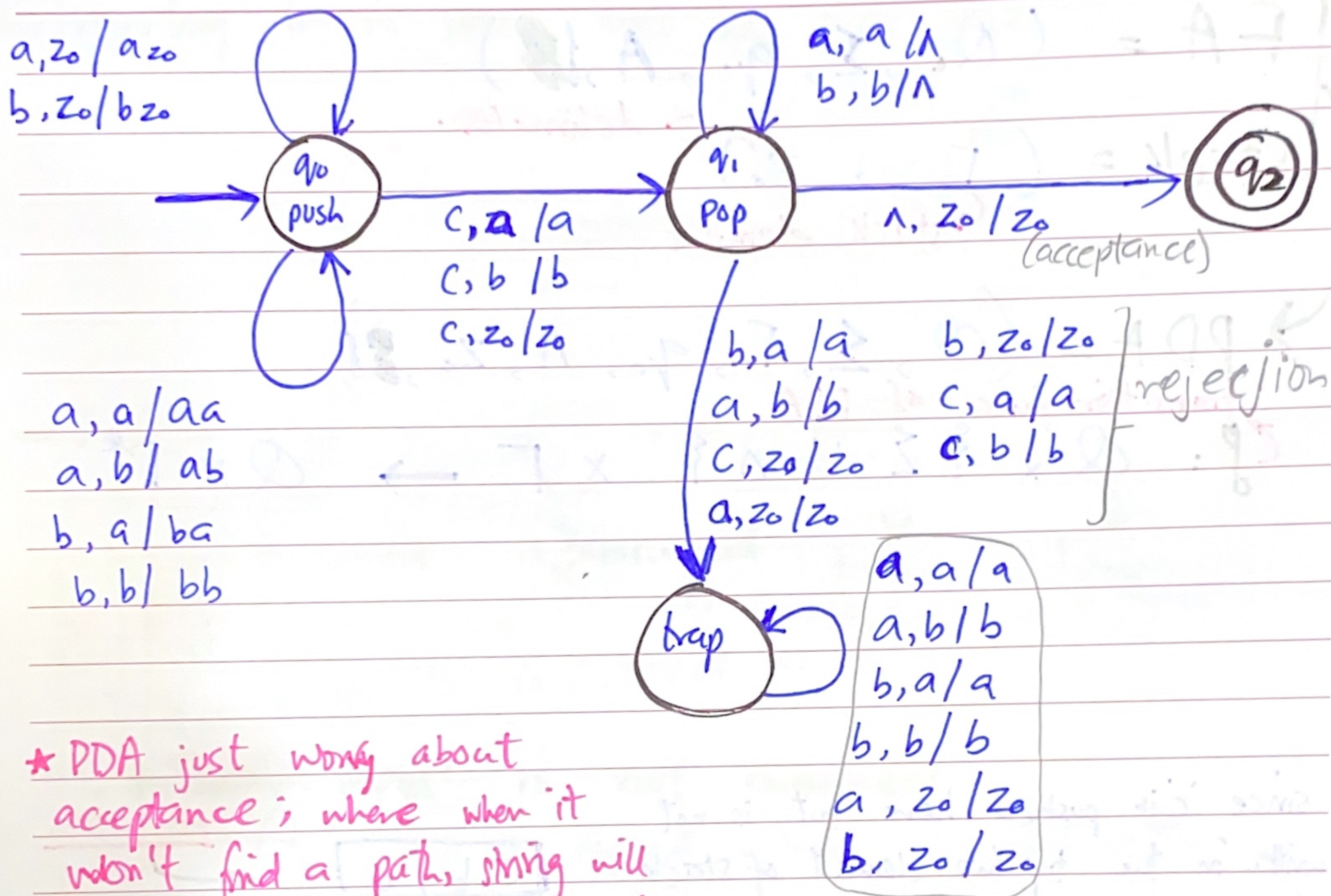
$y^r = \text{reverse of } y$ Palindrome

$a^1 b^2 b^3 c^4 b^5 b^6 a^7$

1	a, Z ₀ / a Z ₀	Push		
2	b, a / ba			
3	b, b / bb			
4	c, b / b	// no action since 'c' not in top two		
5	b, b / Λ	Pop		
6	b, b / Λ			
7	a, a / Λ			

PDA of Palindrome .

day / date:



★ PDA just worry about acceptance; where when it won't find a path, string will automatically be rejected.

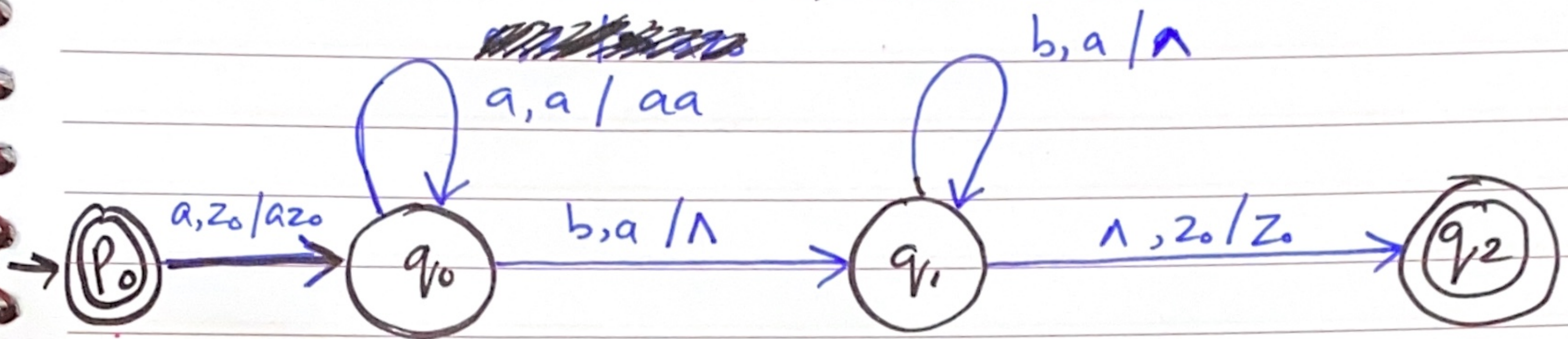
∴ we don't have to make rejection arcs & trap.

• Testing $abcbac$: (By traversing PDA diagram)

State	Input	Top of Stack
q_0	$abcbac$	z_0
q_0	$bcbac$	az_0
q_0	cba	baz_0
q_1	ba	baz_0
q_1	a	az_0
q_1	Λ	z_0
q_2	Λ	z_0

since final state in answer ∴ $abcbac$ accepted.

Q: $\{u \mid u = a^n b^n, \text{ where } n \geq 0\}$.



p_0 is dummy state which is initial & final & is made since Λ is acceptable (as $n \geq 0$). If we made ' q_0 ' initial & final state; then string of only a's will be acceptable whereas we want $n(a) = n(b)$.

• Testing 'ab'

(state, input, TOS)

p_0	ab	z_0
q_0	b	az_0
q_1	Λ	z_0
q_2	Λ	z_0

→ accepted as ' q_2 ' is final.